MMCOE, PUNE

STUDY REPORT

ETHERNET COMMUNICATION

Ethernet Fundamentals, Network Protocols and Linux Networking

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1. Introduction to Ethernet Communication

Ethernet is one of the most widely used technologies for wired Local Area Networks (LANs). It allows computers, servers, switches and other network devices to exchange data over a communication medium. Ethernet mainly operates at the Physical Layer and Data Link Layer of the OSI model.

Ethernet defines how data is formatted into frames, how devices are identified using MAC addresses and how data is transmitted across a local network. Modern Ethernet networks commonly use twisted-pair cables, fibre-optic cables and switches.

2. Ethernet Frame Structure

An Ethernet frame is the basic unit of data transmission at the Ethernet Data Link Layer. It contains addressing information, protocol identification, payload data and an error-checking field.

Field	Size	Purpose
Destination MAC Address	6 bytes	Identifies the receiving device.
Source MAC Address	6 bytes	Identifies the sending device.
EtherType	2 bytes	Identifies the encapsulated protocol, such as IPv4 or ARP.
Payload	46–1500 bytes	Carries the upper-layer data, with padding when required.
Frame Check Sequence (FCS)	4 bytes	Uses CRC to detect transmission errors.

For a standard untagged Ethernet frame, the frame size is normally 64 to 1518 bytes, excluding the preamble and Start Frame Delimiter. The preamble is 7 bytes and the Start Frame Delimiter (SFD) is 1 byte. A VLAN tag adds 4 bytes to the frame.

3. MAC Addressing

A Media Access Control (MAC) address is a 48-bit hardware address used to identify a network interface at the Data Link Layer. It is generally represented as six hexadecimal pairs, for example, 00:15:5D:9B:50:C1.

The first 24 bits generally identify the manufacturer through an assigned Organizationally Unique Identifier (OUI), while the remaining bits identify the interface within that assignment. MAC addresses are used for communication within the local Ethernet network.

4. IPv4 Addressing

Internet Protocol version 4 (IPv4) operates at the Network Layer. It uses a 32-bit logical address, usually written as four decimal numbers separated by dots, such as 192.168.1.10.

An IPv4 address identifies a device interface on an IP network. A subnet mask or prefix length identifies the network portion and host portion of the address. IP packets can be forwarded between different networks using routers.

5. Address Resolution Protocol (ARP)

Address Resolution Protocol (ARP) is used in IPv4 local networks to find the MAC address associated with a known IPv4 address. When a device needs to send a packet to another device on the local network, it checks its ARP cache.

If the required mapping is not available, the device broadcasts an ARP Request asking which device owns the target IPv4 address. The device with that address normally responds with an ARP Reply containing its MAC address. The sender can then use this mapping to construct an Ethernet frame.

ARP is important because IPv4 provides logical addressing, while Ethernet frame delivery on a local network requires MAC addressing.

6. Internet Control Message Protocol (ICMP)

Internet Control Message Protocol (ICMP) is used with IP for network diagnostics and reporting network-related errors. It does not transport application data like TCP or UDP.

The ping utility commonly uses ICMP Echo Request and Echo Reply messages to check whether a destination is reachable and to measure round-trip time. An Echo Request has type 8 and code 0, while an Echo Reply has type 0 and code 0 in IPv4.

7. Transmission Control Protocol (TCP)

Transmission Control Protocol (TCP) is a connection-oriented transport-layer protocol. It provides reliable, ordered delivery of data between applications. TCP uses sequence numbers, acknowledgements, retransmission and flow control to support reliable communication.

Before normal data transfer, TCP establishes a connection using a three-way handshake:

1. **SYN:** The client requests a connection.
2. **SYN-ACK:** The server acknowledges the request and sends its own synchronization request.
3. **ACK:** The client acknowledges the server, and the connection is established.

TCP is commonly used by applications such as web browsing, file transfer and email. HTTP communication can use TCP port 80, while HTTPS commonly uses TCP port 443.

8. User Datagram Protocol (UDP)

User Datagram Protocol (UDP) is a connectionless transport-layer protocol. It sends independent datagrams without establishing a connection before transmission. UDP has a smaller header and lower protocol overhead than TCP.

UDP does not guarantee delivery, ordering or retransmission of lost datagrams. Applications that use UDP must handle any reliability requirements themselves when needed. UDP is commonly used for DNS queries, streaming, online communication and real-time applications.

DNS commonly uses UDP port 53 for standard queries, although TCP is also used in specific situations.

9. Client-Server Communication

Client-server communication is a networking model in which a client requests a service and a server provides a response. A client can be a browser or another application, while a server hosts the requested resource or service.

For example, when a user opens a website, the client communicates with a web server. The

communication may involve DNS resolution, TCP connection establishment and HTTP request-response messages.

10. Linux Network Interfaces

Linux represents network adapters as network interfaces. Common interface names include eth0 for an Ethernet interface and lo for the loopback interface. The loopback interface allows a device to communicate with itself.

Network interfaces have properties such as MAC address, IP address, MTU, operational state and packet statistics. Linux networking tools can display and help investigate these properties.

11. Linux Networking Commands

Command	Purpose
`ip addr`	Displays network interfaces and IP addresses.
`ip route`	Displays the IP routing table.
`arp -a`	Displays the ARP cache.
`ping`	Tests network reachability using ICMP.
`ss -tuln`	Displays listening TCP and UDP sockets.
`netstat -tuln`	Displays listening network sockets.
`hostname -I`	Displays the host's assigned IP addresses.
`ip -s link`	Displays network interface statistics.

These commands help investigate network configuration, routing, address resolution, connectivity and socket information in Linux.

12. Conclusion

Ethernet provides the foundation for wired local area network communication. Understanding Ethernet frames, MAC and IP addressing, ARP, ICMP, TCP and UDP helps explain how devices exchange data. Linux networking commands provide practical methods to inspect interfaces, routes, neighbours, connectivity and sockets. These concepts form the theoretical foundation for capturing and analysing network traffic.

13. References

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3. RFC 791 – Internet Protocol.
4. RFC 792 – Internet Control Message Protocol.
5. RFC 9293 – Transmission Control Protocol (TCP).
6. RFC 768 – User Datagram Protocol.
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8. Wireshark Documentation, https://www.wireshark.org/docs/